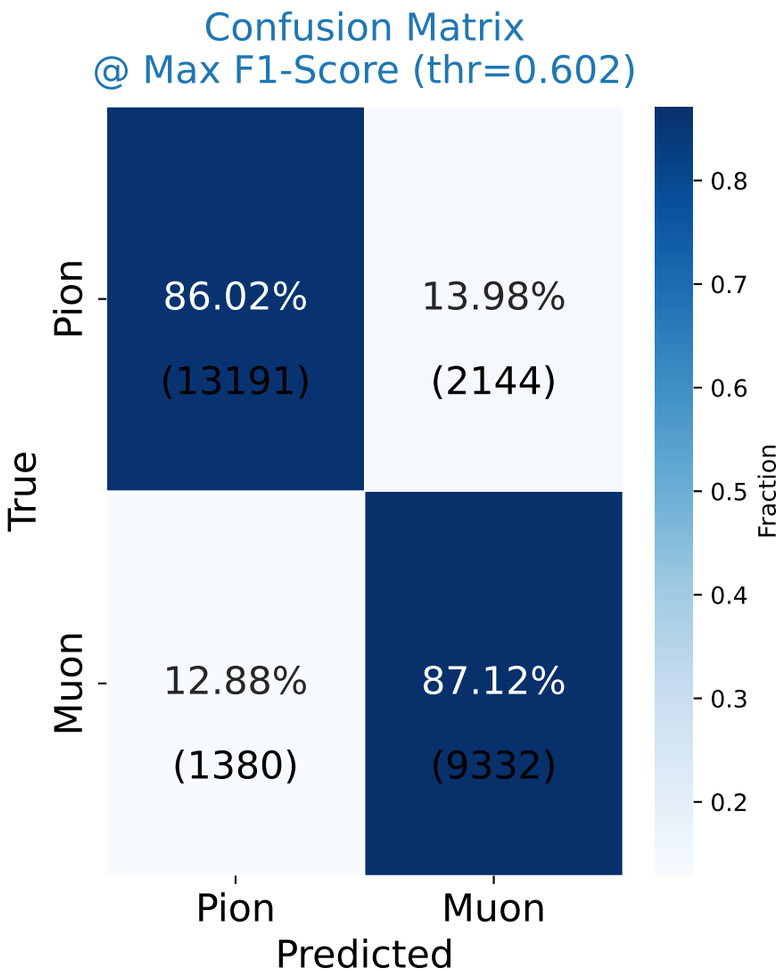
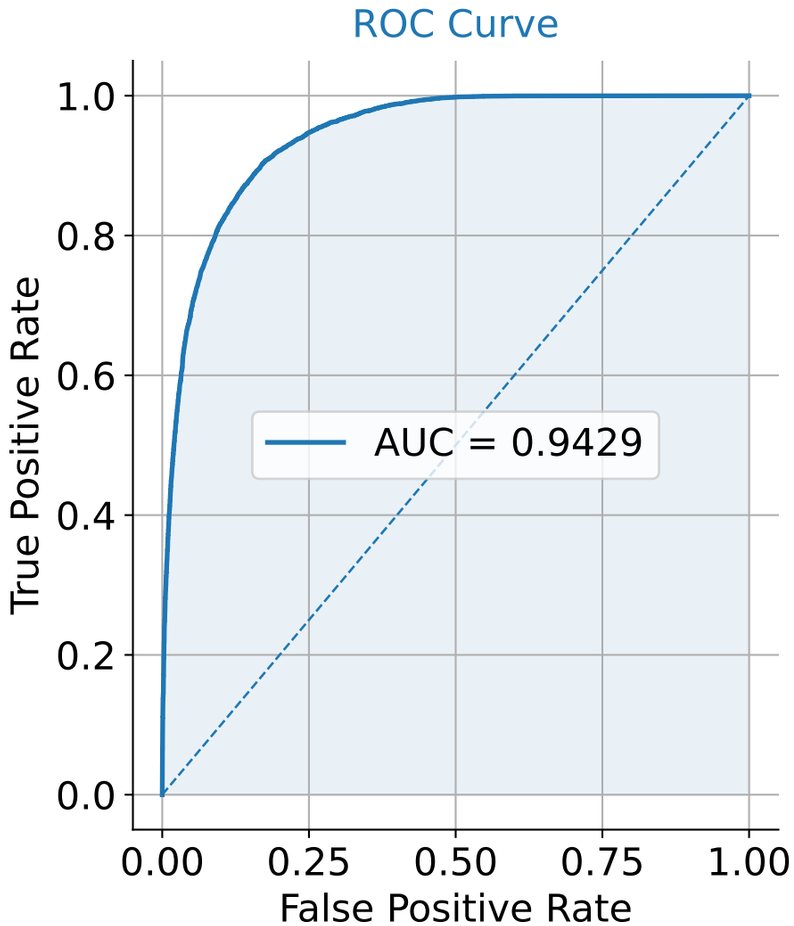
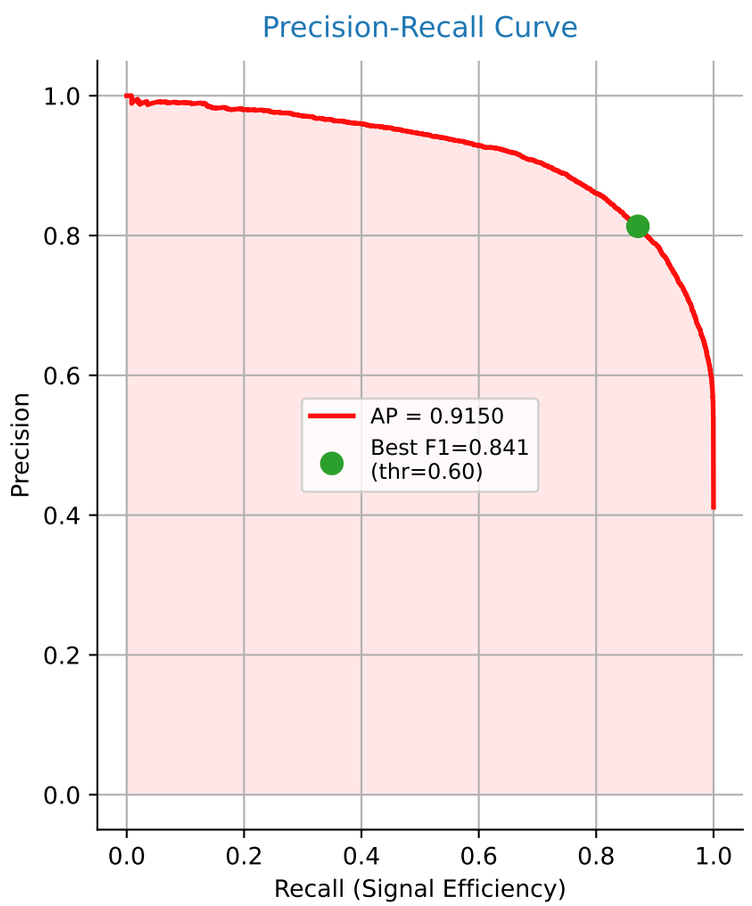
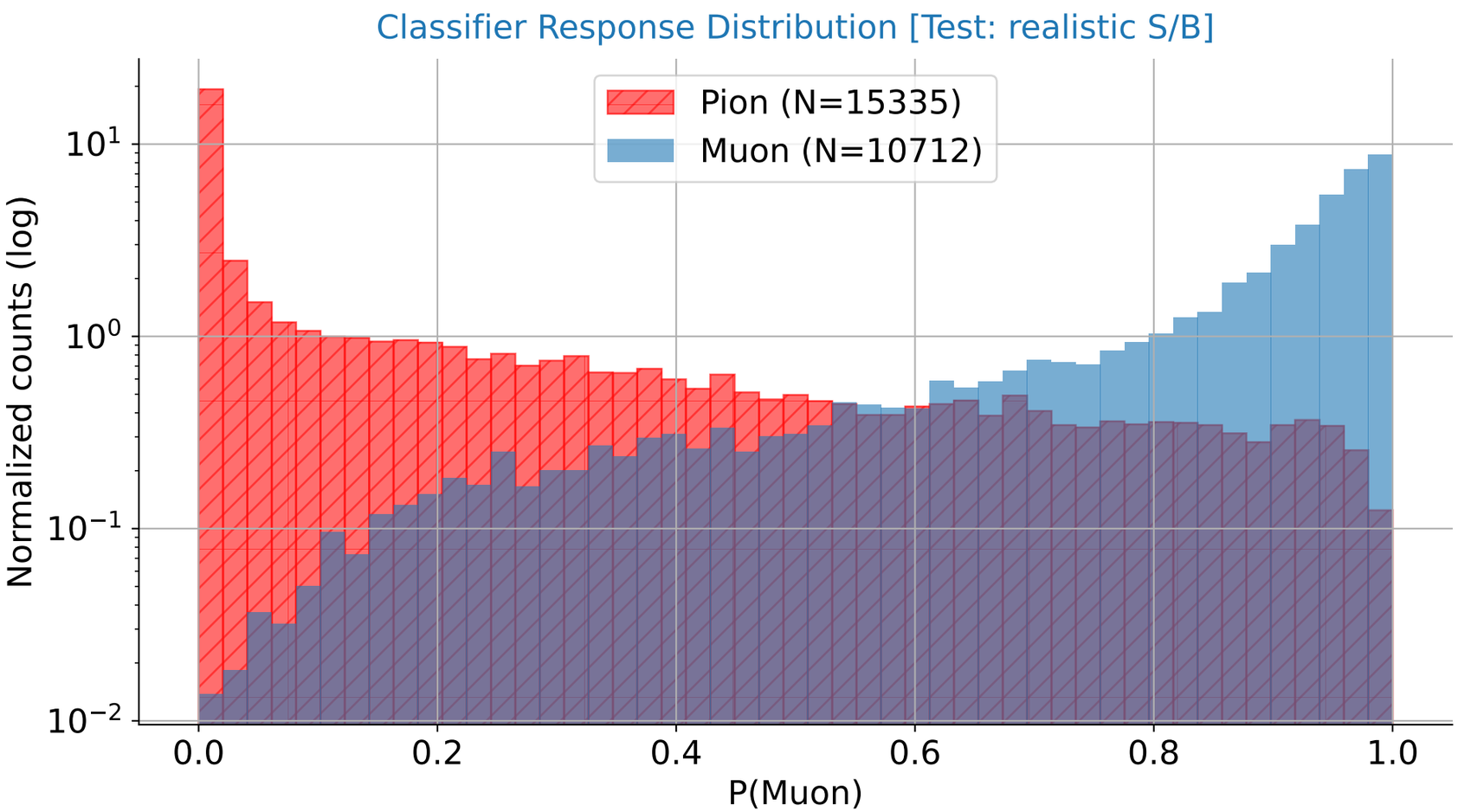
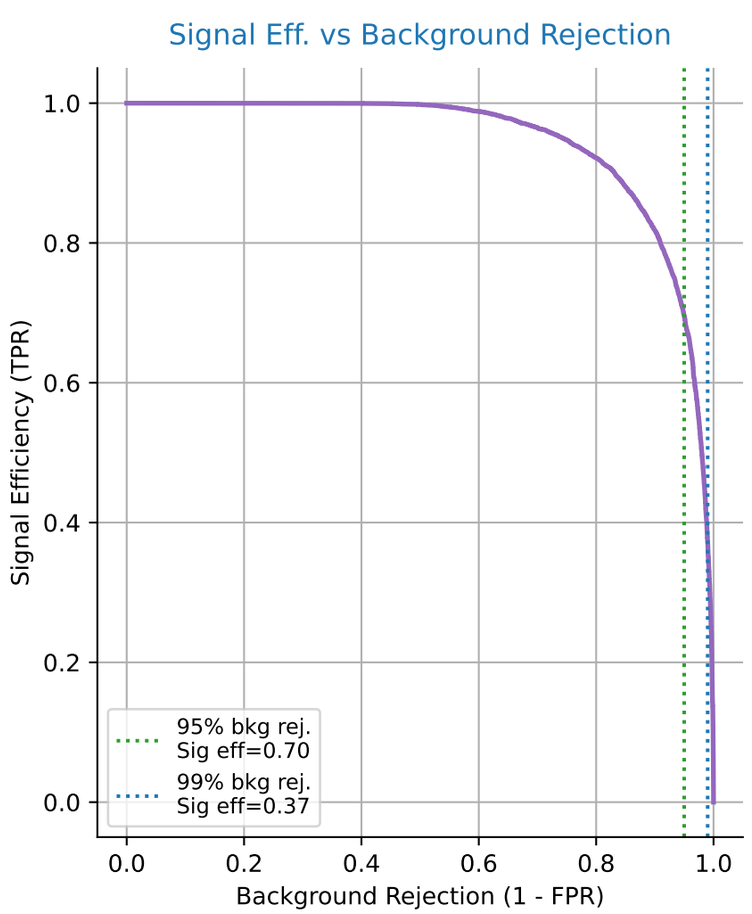
Muon vs Pion ID — XGBoost (Calorimeter + ToF features)
[Train S/B=1:1 | Test S/B=1:1 (realistic, all files)]



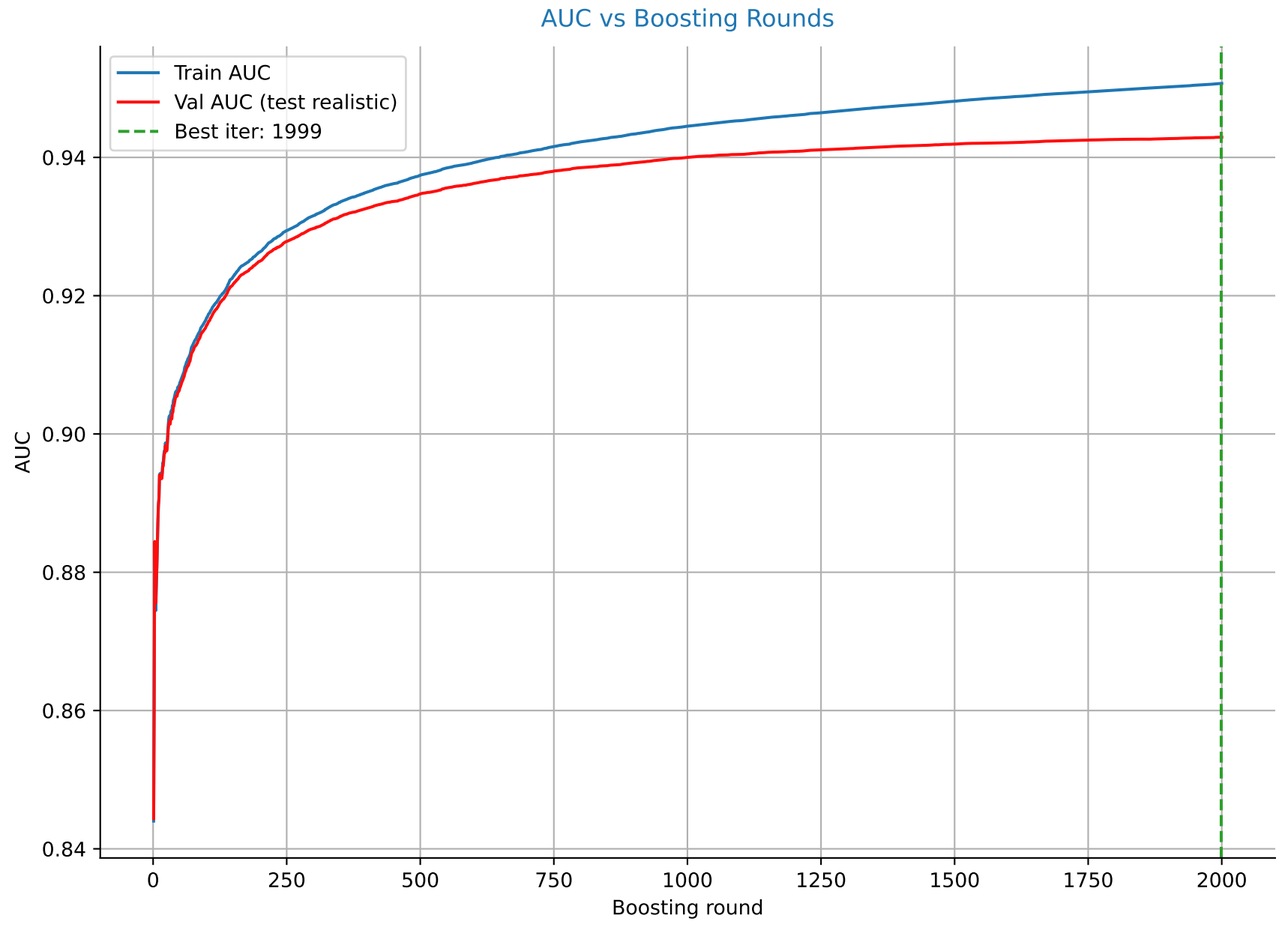


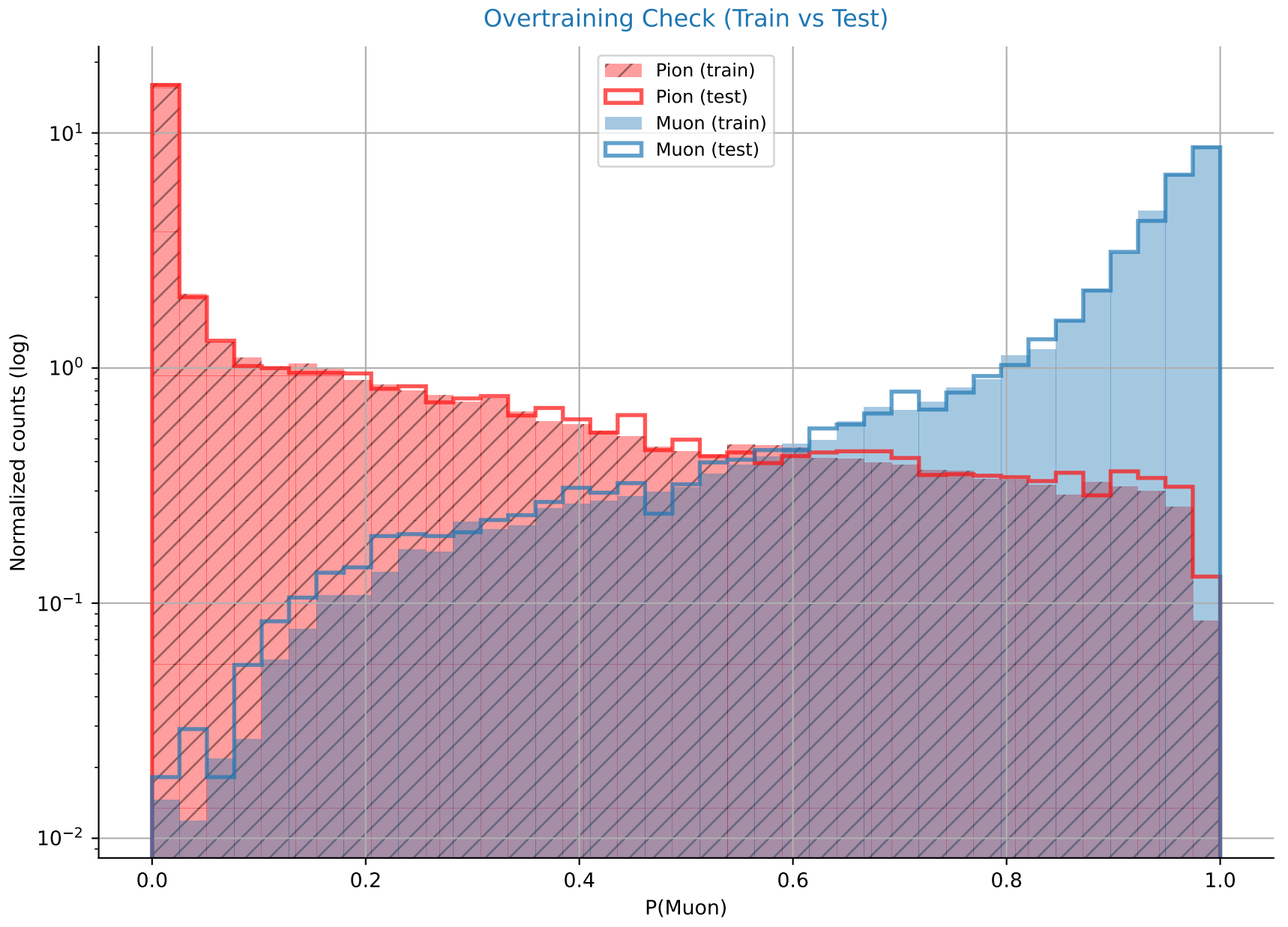






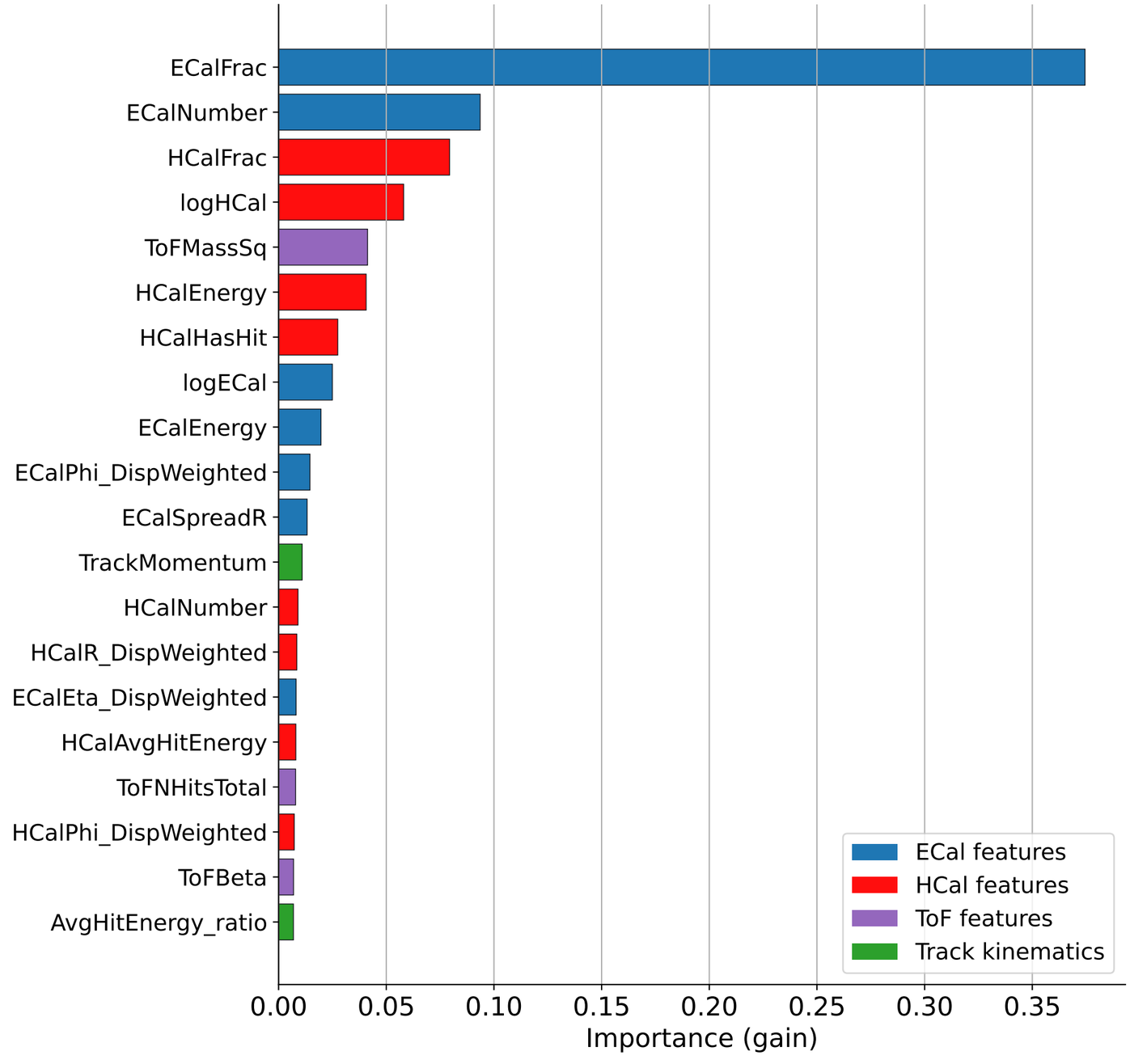
XGBoost Training Diagnostics



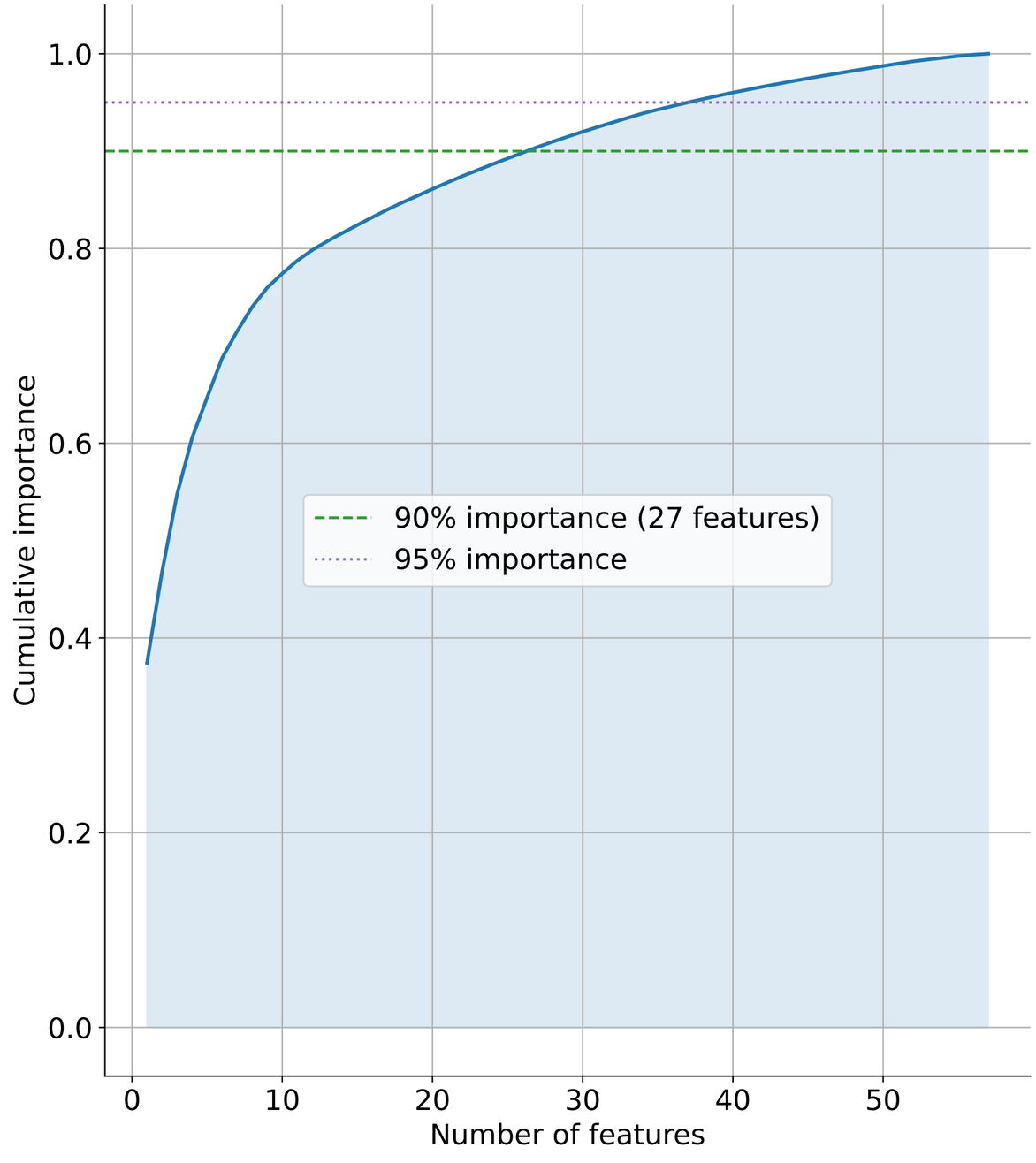


Feature Importances — XGBoost

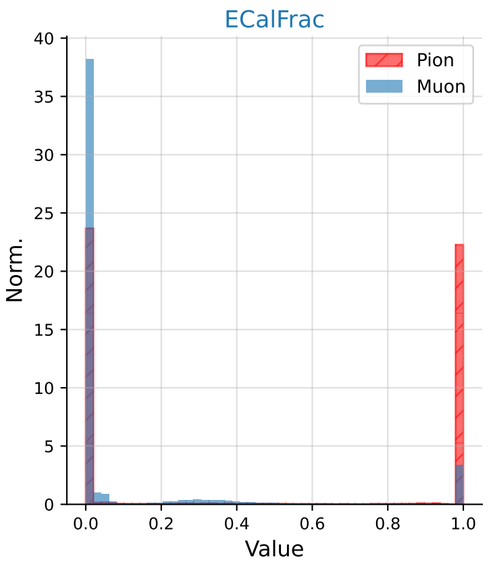
Top 20 Features

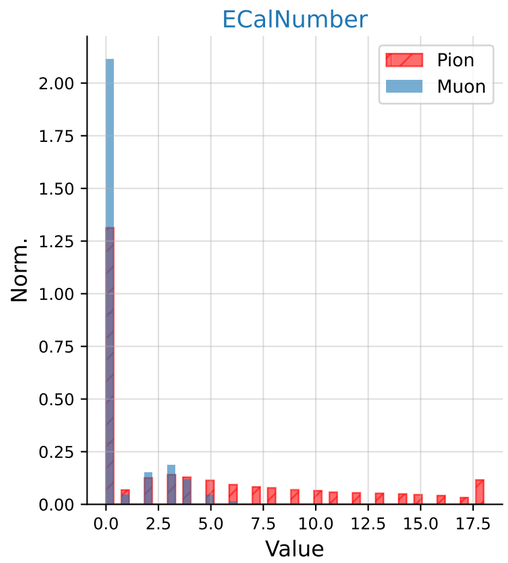


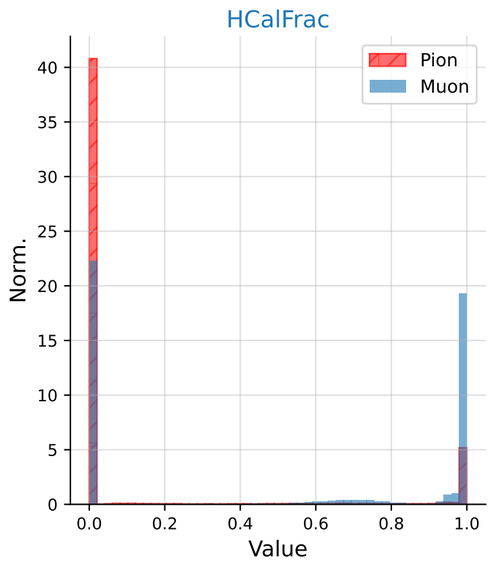
Cumulative Feature Importance

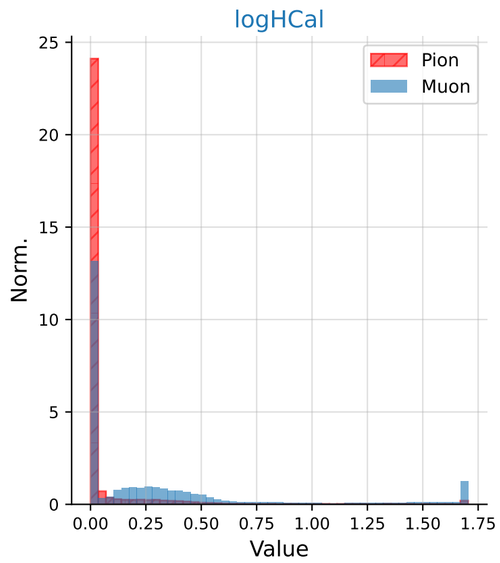


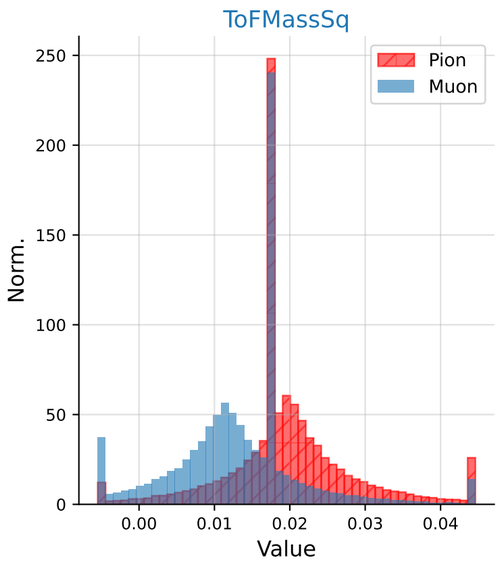
Key Physics Variable Distributions: Muon vs Pion
[full dataset before splitting]

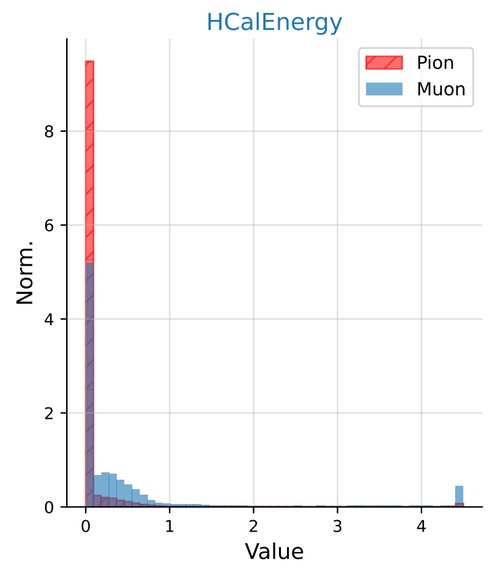


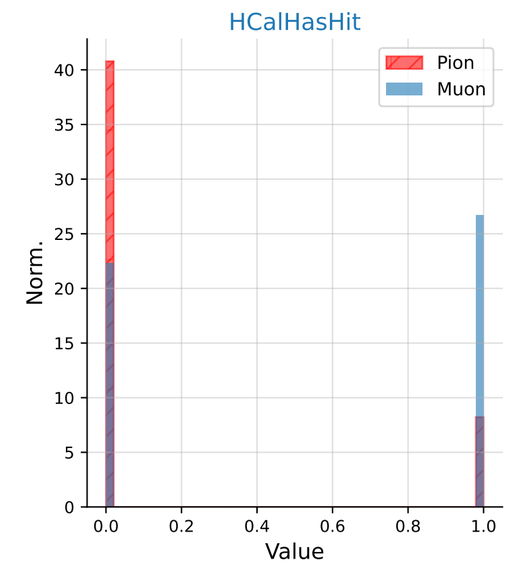


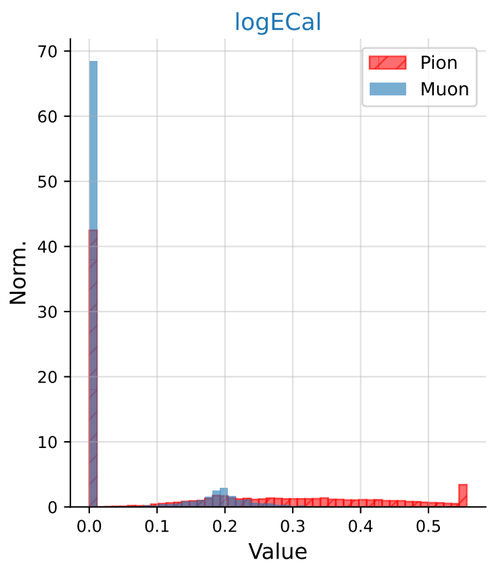


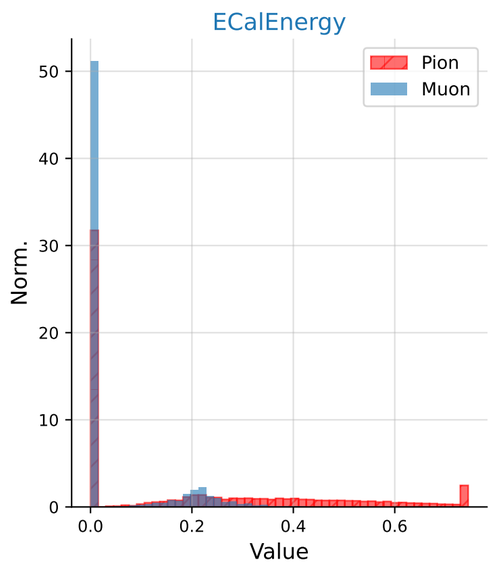


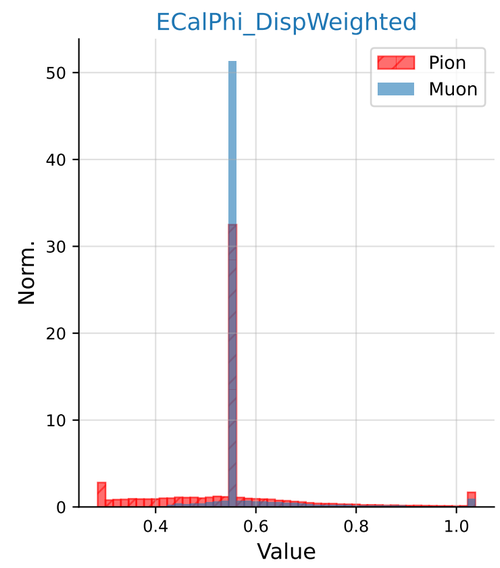


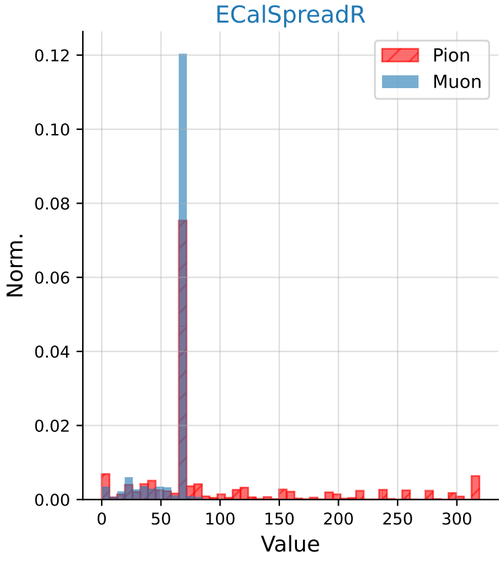


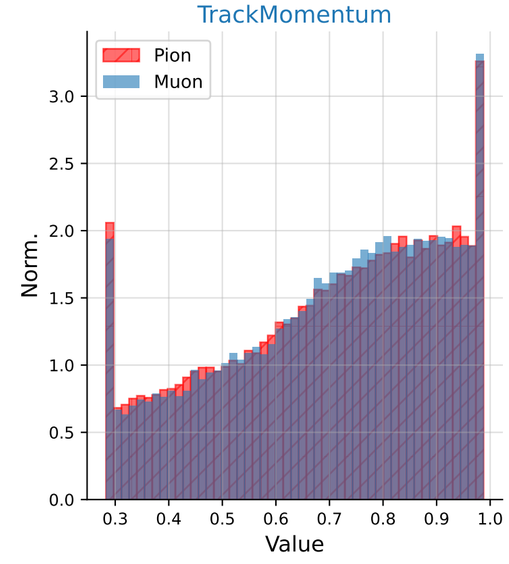


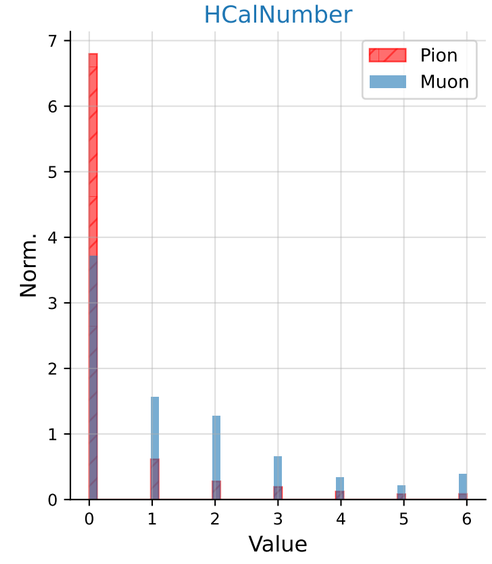


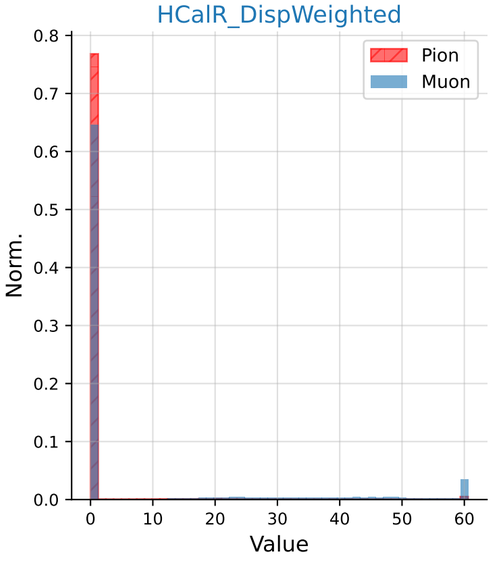


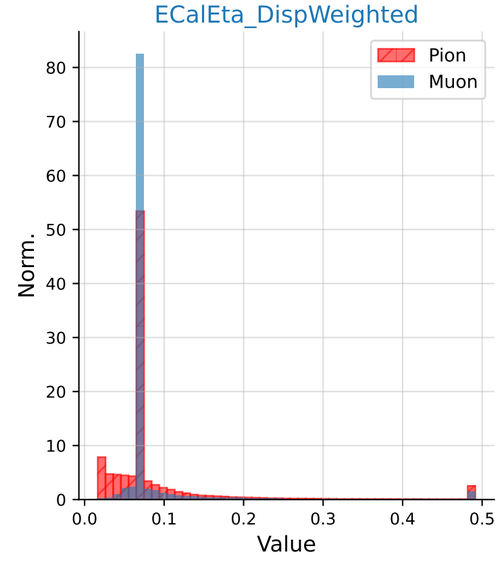






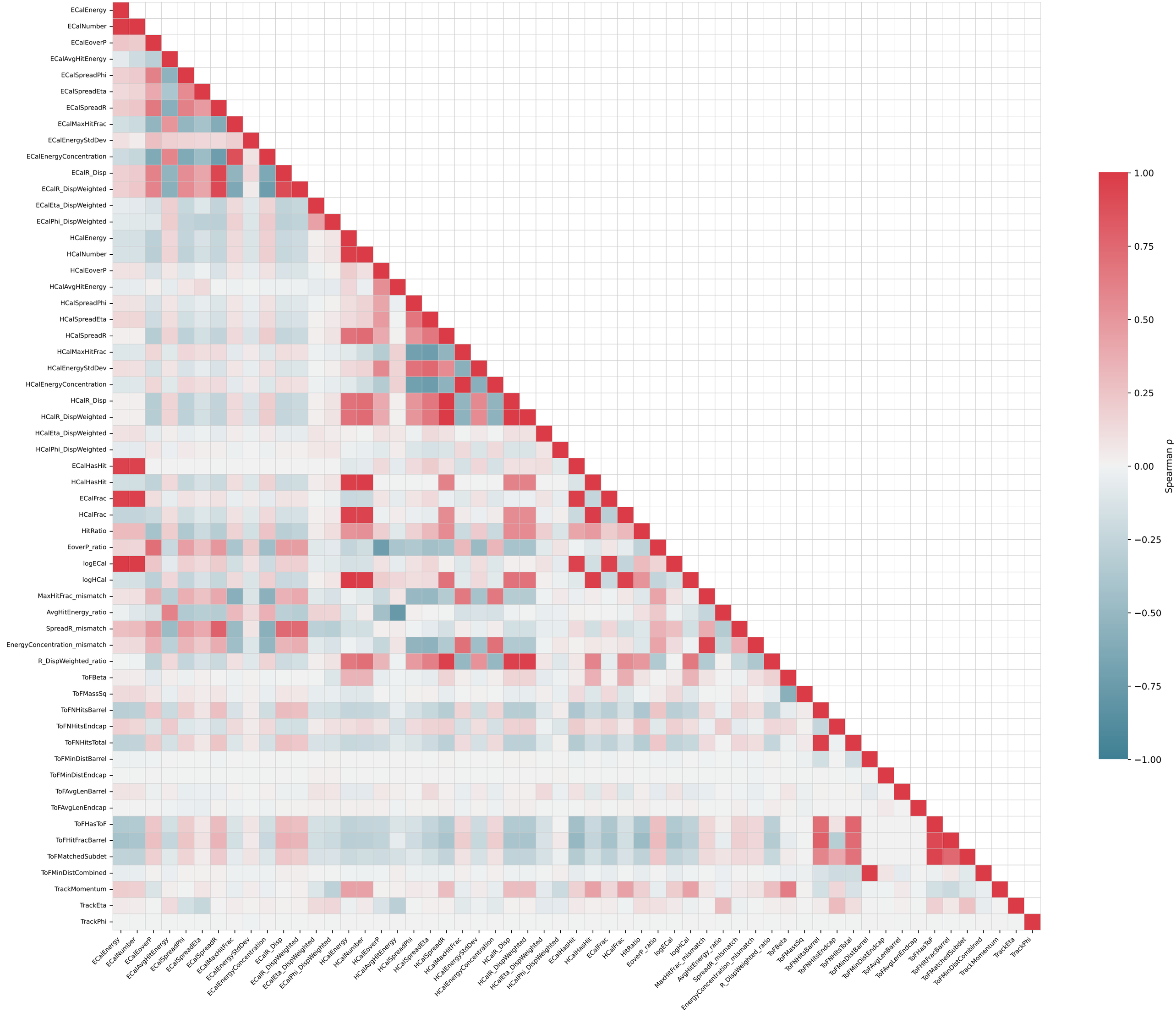






Spearman Correlation Matrix — All Features
(values from the full dataset)

Higher |ρ| means a stronger monotonic relationship between features

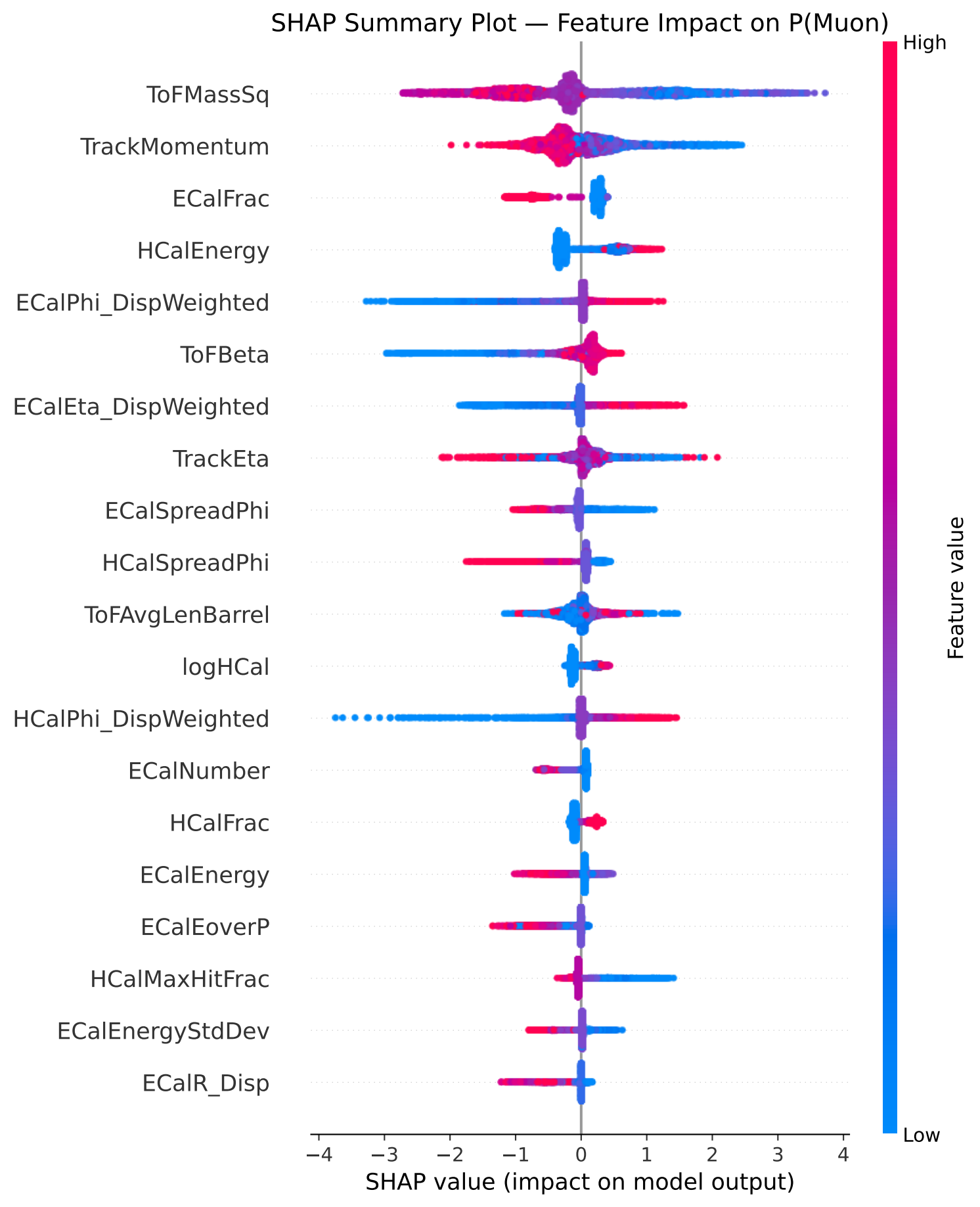


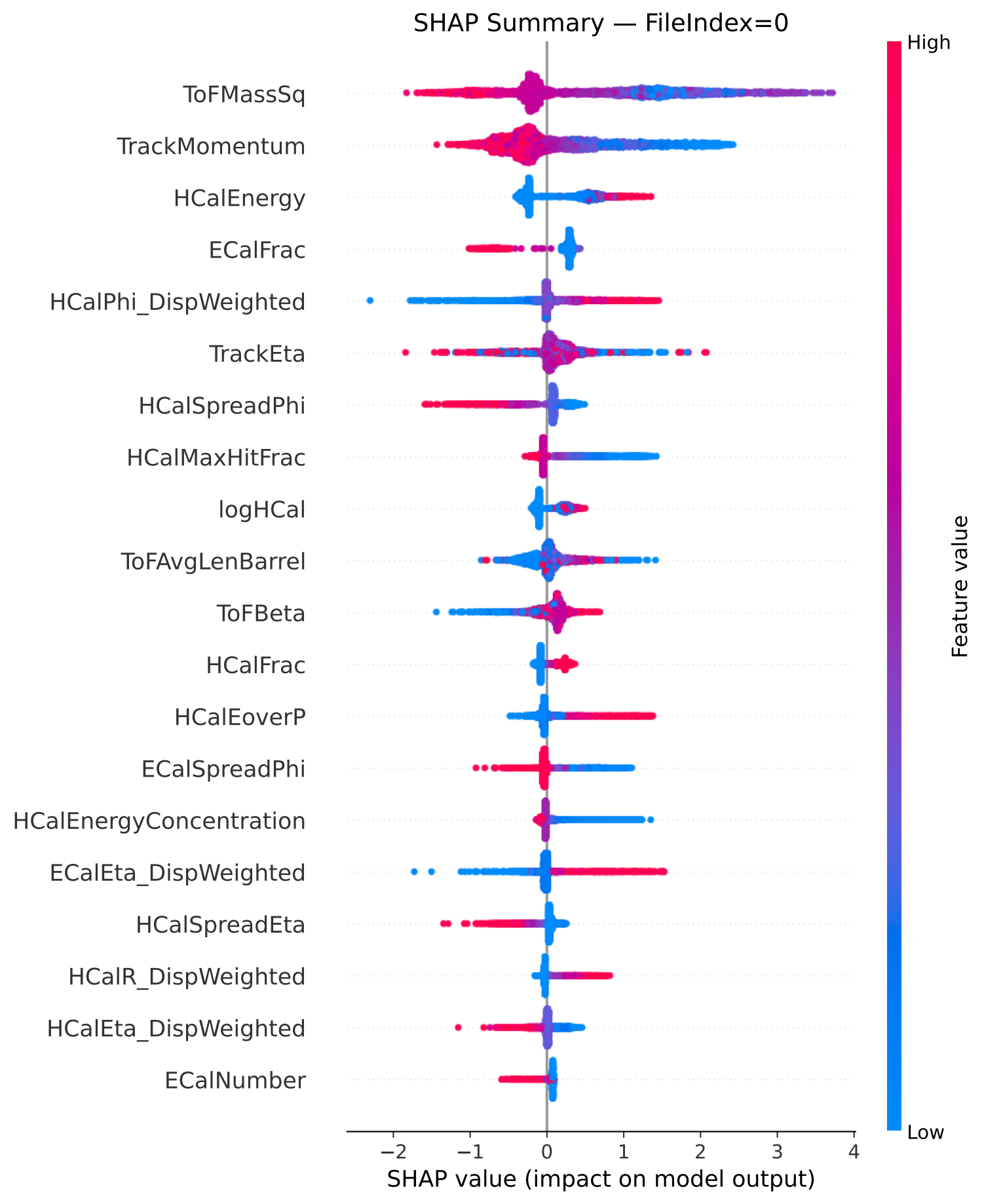
Feature correlation with muon/pion class
Blue = positive (→ muon), Red = negative (→ pion)

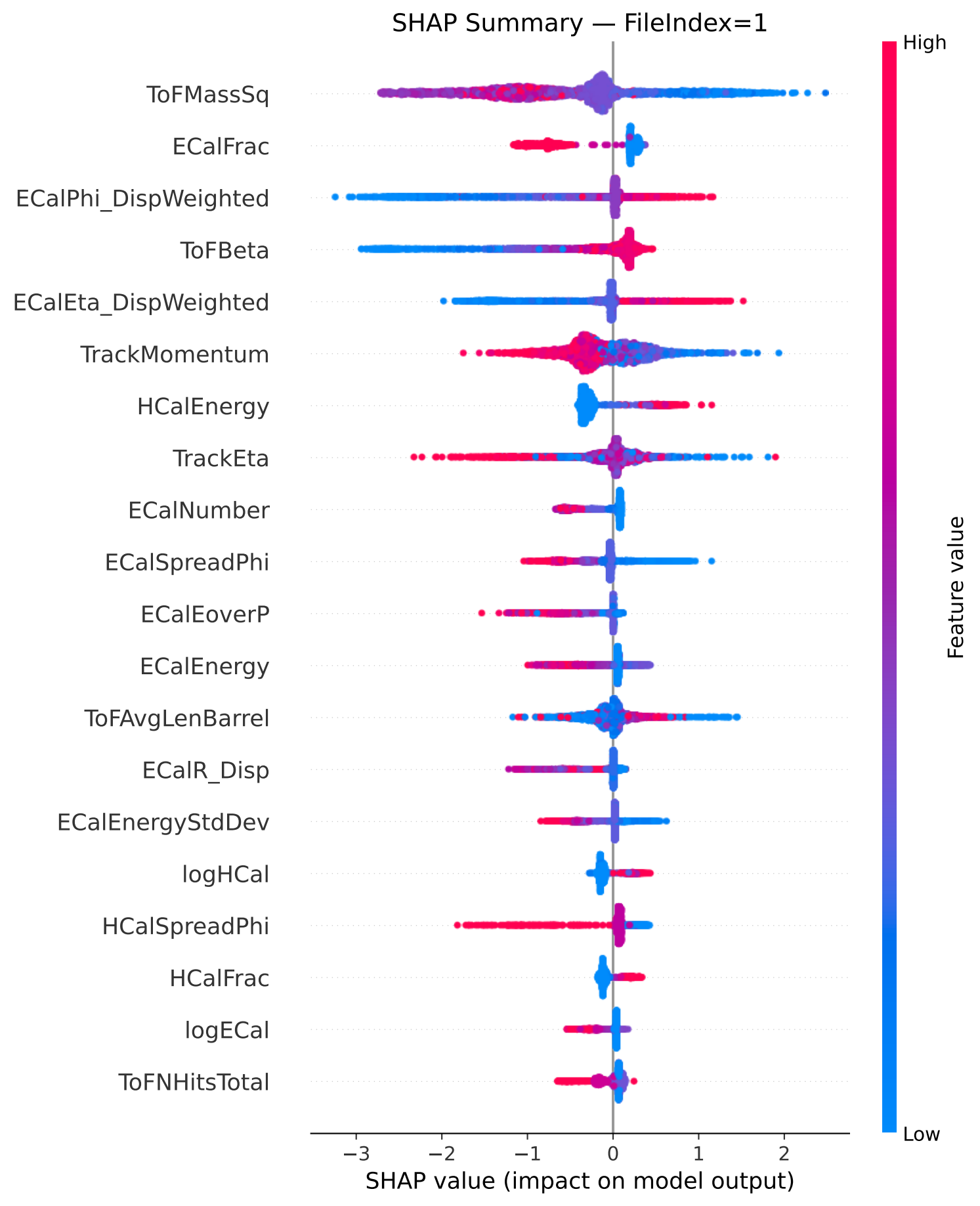
ToFMinDistEndcap
TrackEta
ToFAvgLenBarrel
ToFNHitsEndcap
ToFAvgLenEndcap
TrackMomentum
TrackPhi
ToFMatchedSubdet
ToFHasToF
ToFHitFracBarrel
HCalSpreadPhi
HCalSpreadEta
ToFMinDistCombined
ToFMinDistBarrel
HCalEta_DispWeighted
AvgHitEnergy_ratio
HCalEnergyConcentration
HCalMaxHitFrac
MaxHitFrac_mismatch
HCalEnergyStdDev
EnergyConcentration_mismatch
ToFNHitsBarrel
ToFNHitsTotal
HCalPhi_DispWeighted
ECalEta_DispWeighted
HCalEoverP
HCalAvgHitEnergy
ECalMaxHitFrac
ECalEnergyStdDev
ECalSpreadEta
ECalAvgHitEnergy
ECalPhi_DispWeighted
HitRatio
ECalEnergyConcentration
SpreadR_mismatch
ECalR_DispWeighted
ECalR_Disp
EoverP_ratio
ECalSpreadPhi
ECalSpreadR
ToFBeta
ECalEoverP
R_DispWeighted_ratio
HCalR_Disp
HCalSpreadR
HCalR_DispWeighted
ECalHasHit
logECal
ECalEnergy
ECalNumber
ECalFrac
HCalNumber
HCalHasHit
ToFMassSq
HCalFrac
HCalEnergy
logHCal

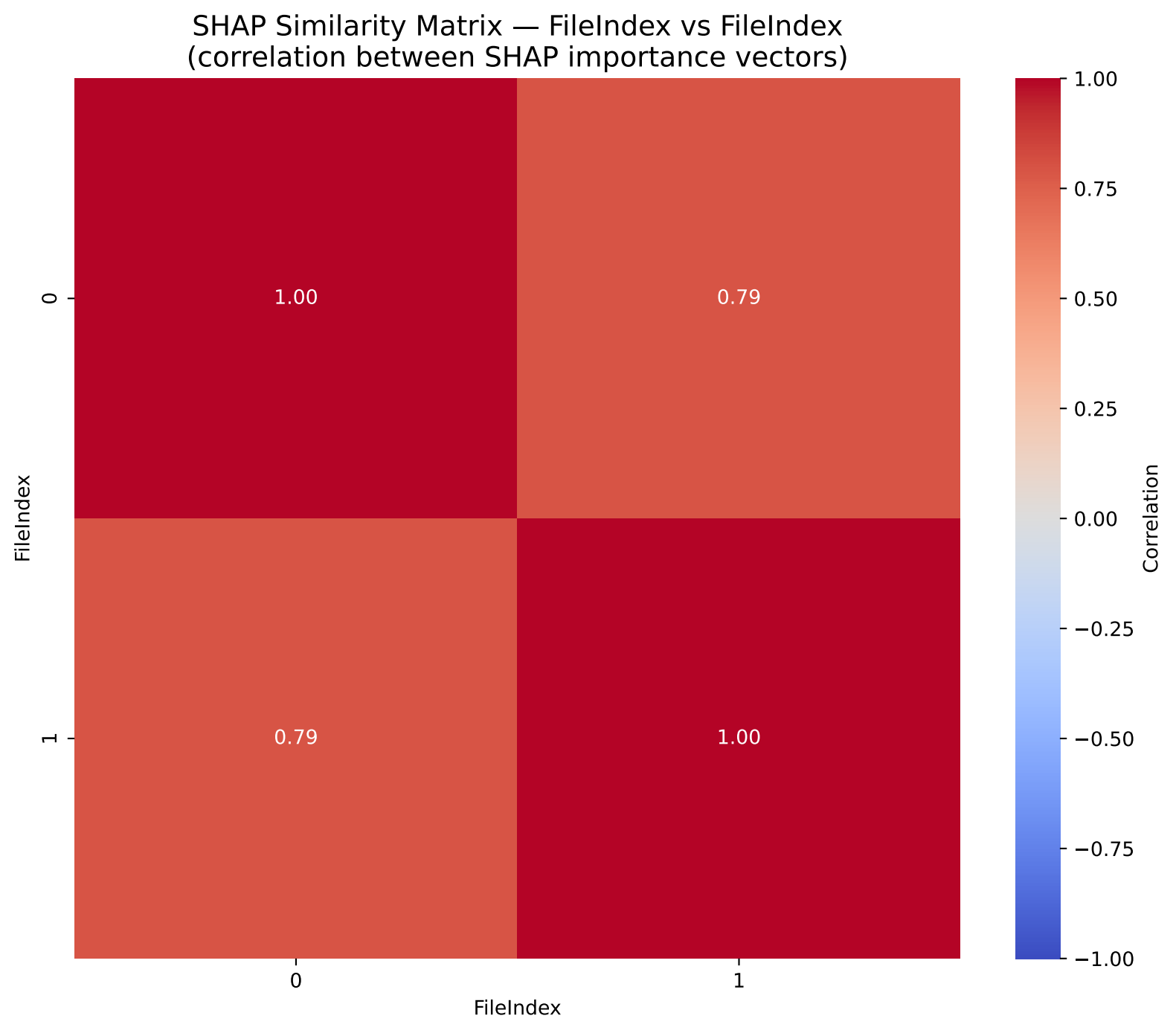
−0.4 −0.3 −0.2 −0.1 0.0 0.1 0.2 0.3 0.4

Spearman ρ with the label (IsMuon)

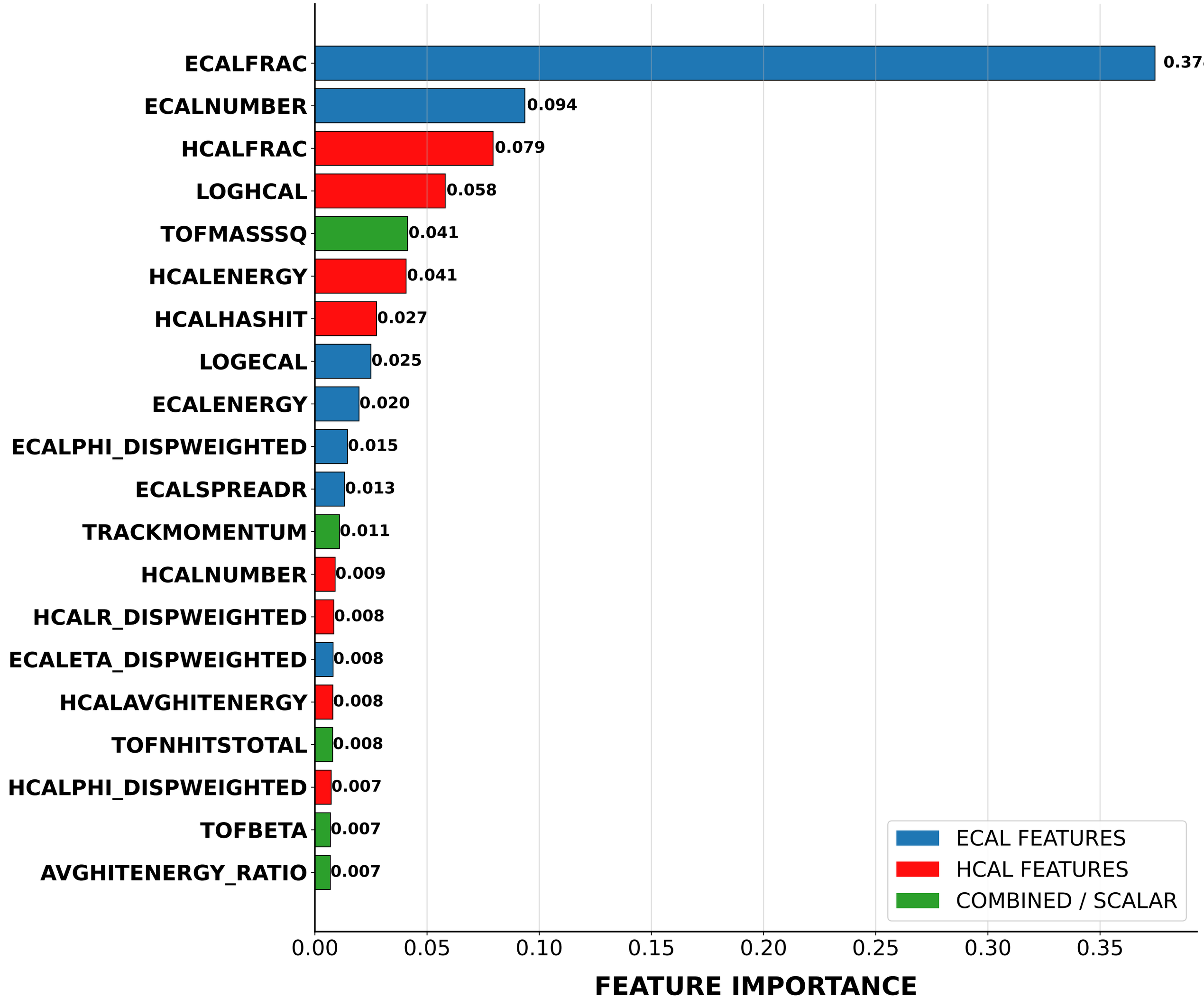

SHAP Summary Plot — Feature Impact on P(Muon)
ToFMassSq
TrackMomentum
ECalFrac
HCalEnergy
ECalPhi_DispWeighted
ToFBeta
ECalEta_DispWeighted
TrackEta
ECalSpreadPhi
HCalSpreadPhi
ToFAvgLenBarrel
logHCal
HCalPhi_DispWeighted
ECalNumber
HCalFrac
ECalEnergy
ECalEoverP
HCalMaxHitFrac
ECalEnergyStdDev
ECalR_Disp
−4
−3
−2
−1
0
1
2
3
4
SHAP value (impact on model output)
High
Feature value
Low


SHAP Summary — FileIndex=0
ToFMassSq
TrackMomentum
HCalEnergy
ECalFrac
HCalPhi_DispWeighted
TrackEta
HCalSpreadPhi
HCalMaxHitFrac
logHCal
ToFAvgLenBarrel
ToFBeta
HCalFrac
HCalEoverP
ECalSpreadPhi
HCalEnergyConcentration
ECalEta_DispWeighted
HCalSpreadEta
HCalR_DispWeighted
HCalEta_DispWeighted
ECalNumber
−2
−1
0
1
2
3
4
SHAP value (impact on model output)
High
Low
Feature value

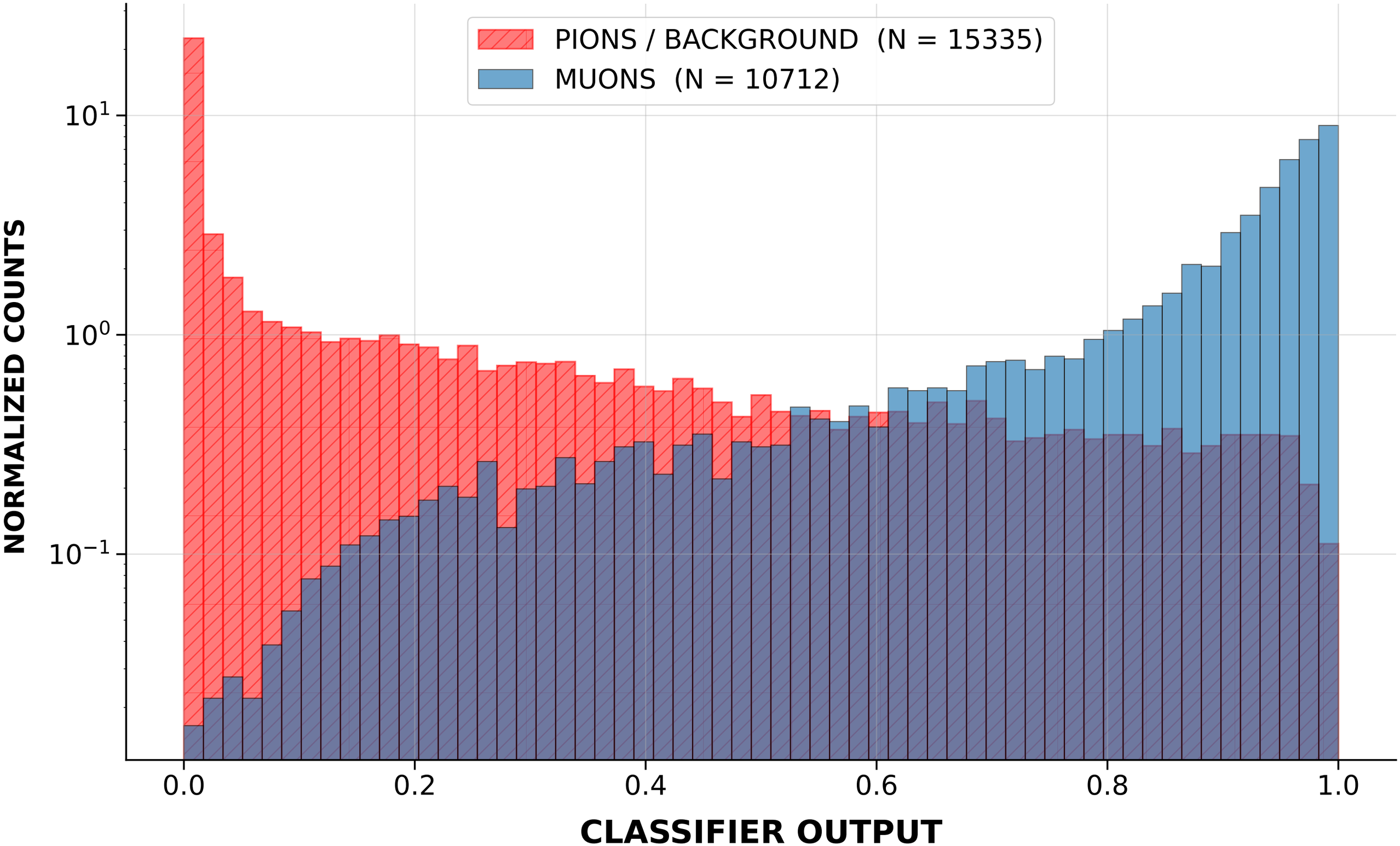

SHAP Summary — FileIndex=1
ToFMassSq
ECalFrac
ECalPhi_DispWeighted
ToFBeta
ECalEta_DispWeighted
TrackMomentum
HCalEnergy
TrackEta
ECalNumber
ECalSpreadPhi
ECalEoverP
ECalEnergy
ToFAvgLenBarrel
ECalR_Disp
ECalEnergyStdDev
logHCal
HCalSpreadPhi
HCalFrac
logECal
ToFNHitsTotal
−3
−2
−1
0
1
2
SHAP value (impact on model output)
High
Feature value
Low


SHAP Similarity Matrix — FileIndex vs FileIndex
(correlation between SHAP importance vectors)
1.00
0.79
0.79
1.00
FileIndex
0
1
FileIndex
0
1
Correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

FEATURE IMPORTANCE — TOP 20



CLASSIFIER SCORE DISTRIBUTION
MUONS vs PIONS



ROC CURVE: ε_μ vs $\varepsilon_{\pi,\text{rej}}$

MUON IDENTIFICATION EFFICIENCY

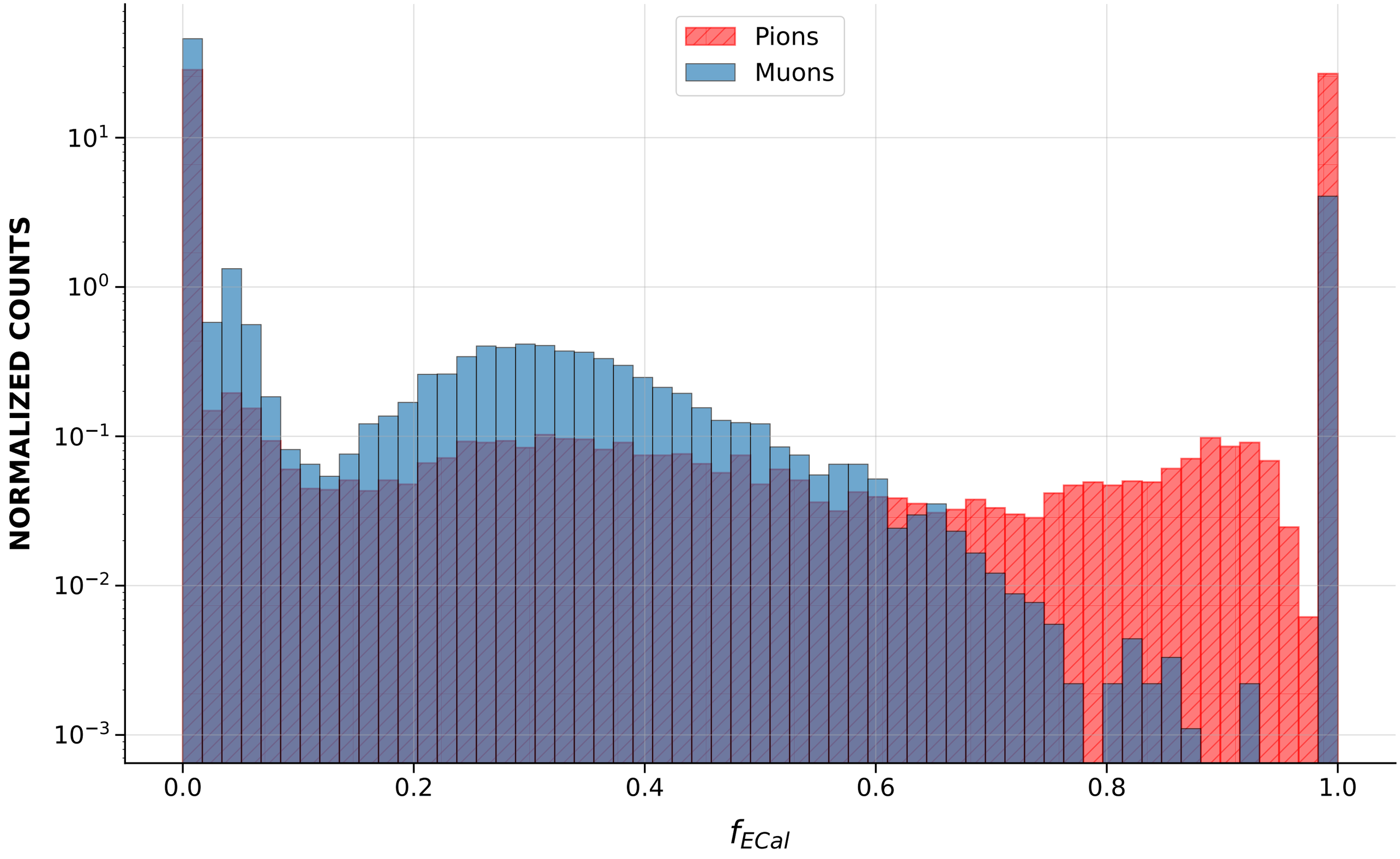
1.0
0.8
0.6
0.4
0.2
0.0

AUC = 0.9429
$\varepsilon_{\pi,rej}$=95% → ε_μ=0.696
$\varepsilon_{\pi,rej}$=99% → ε_μ=0.371

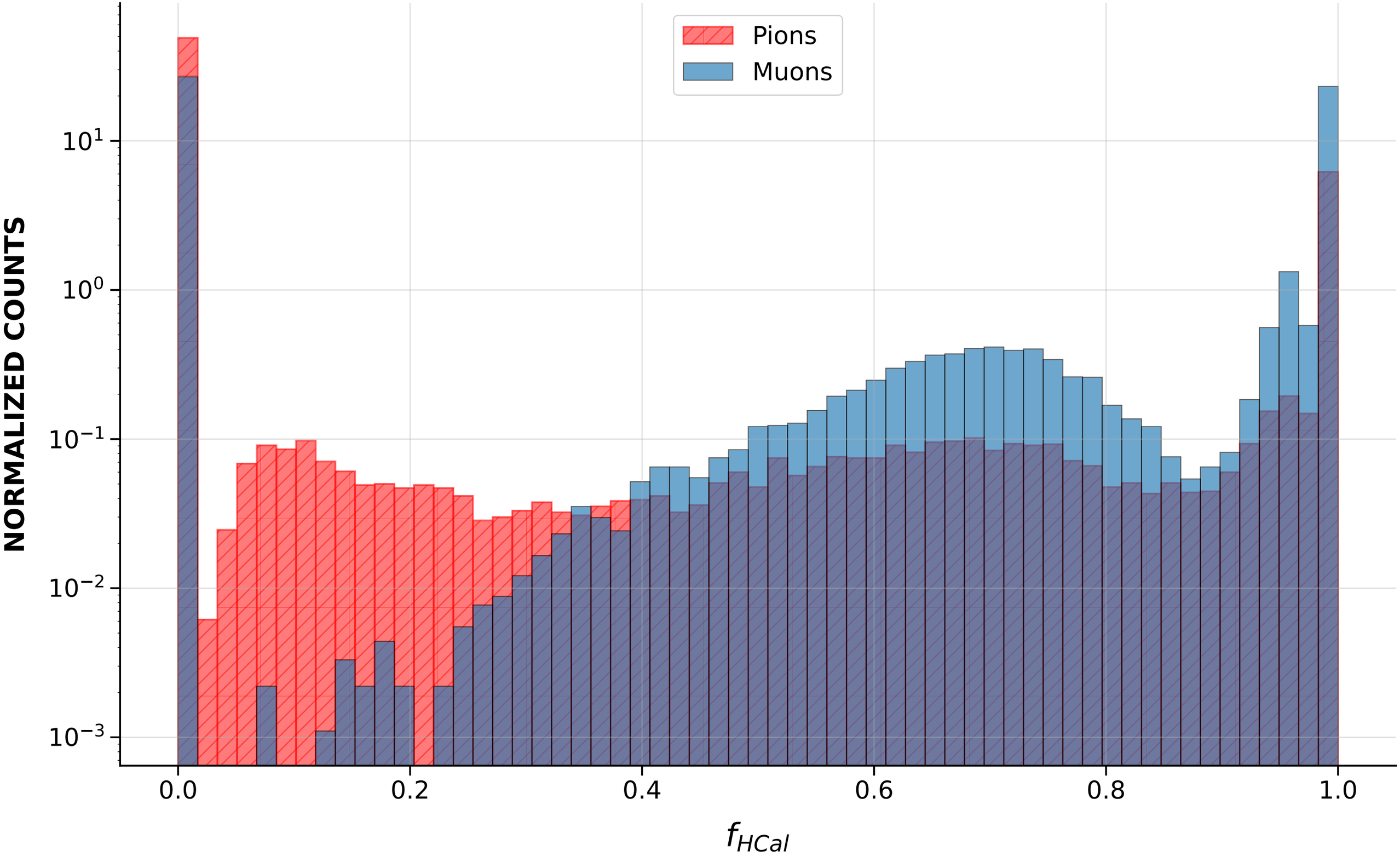
0.0 0.2 0.4 0.6 0.8 1.0

PION REJECTION EFFICIENCY

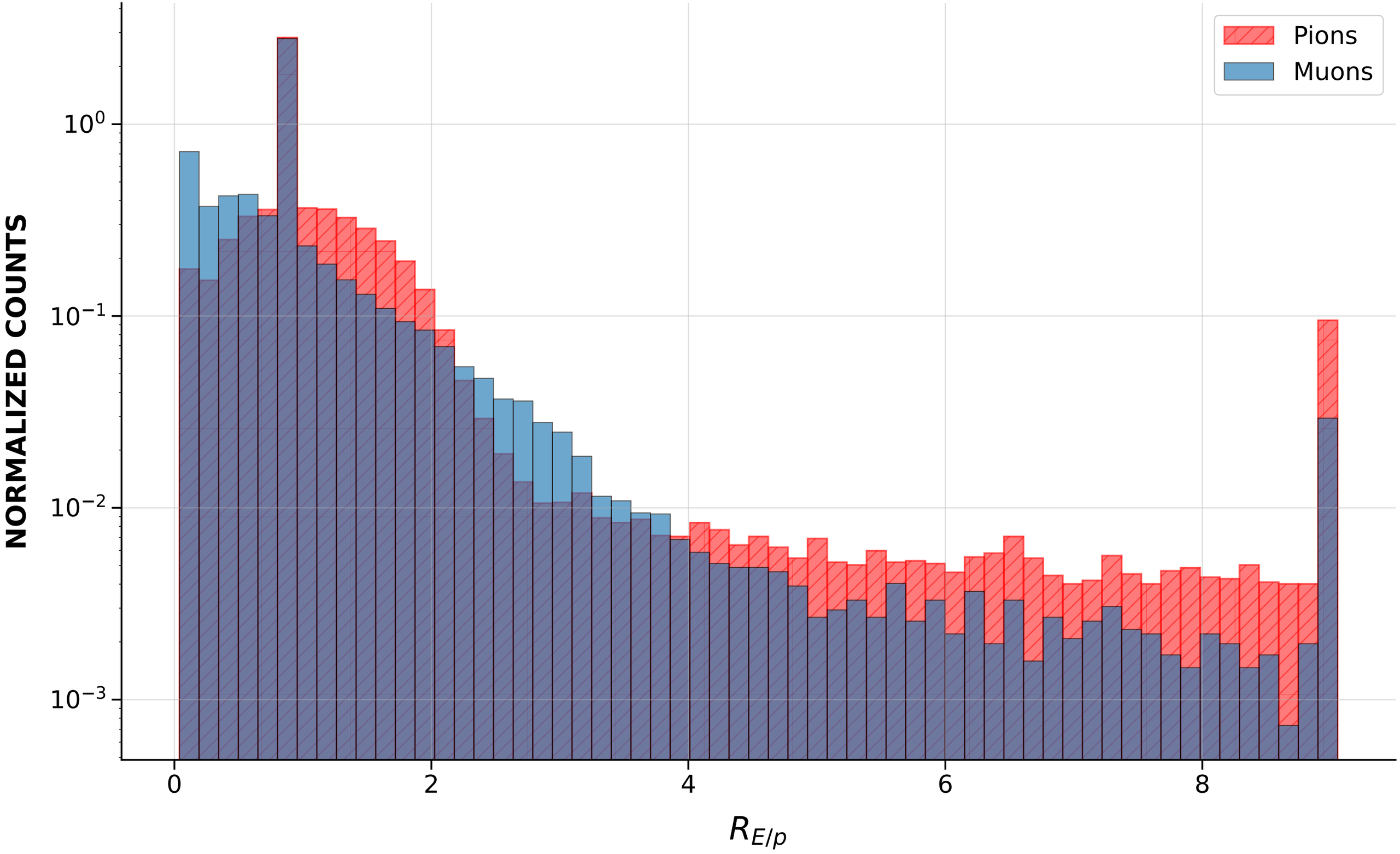
ENGINEERED CALORIMETER FEATURE
ECalFrac



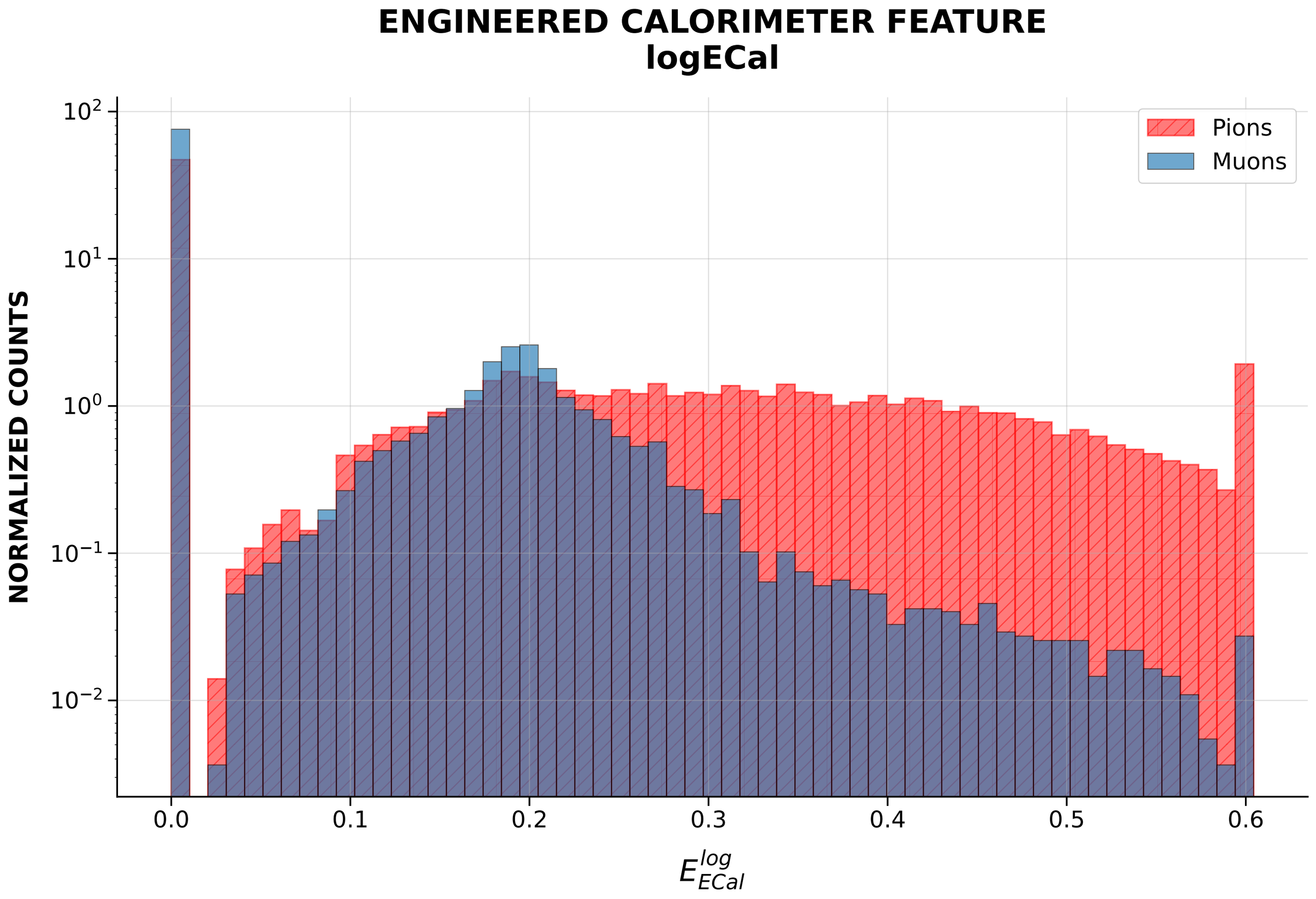
ENGINEERED CALORIMETER FEATURE
HCalFrac



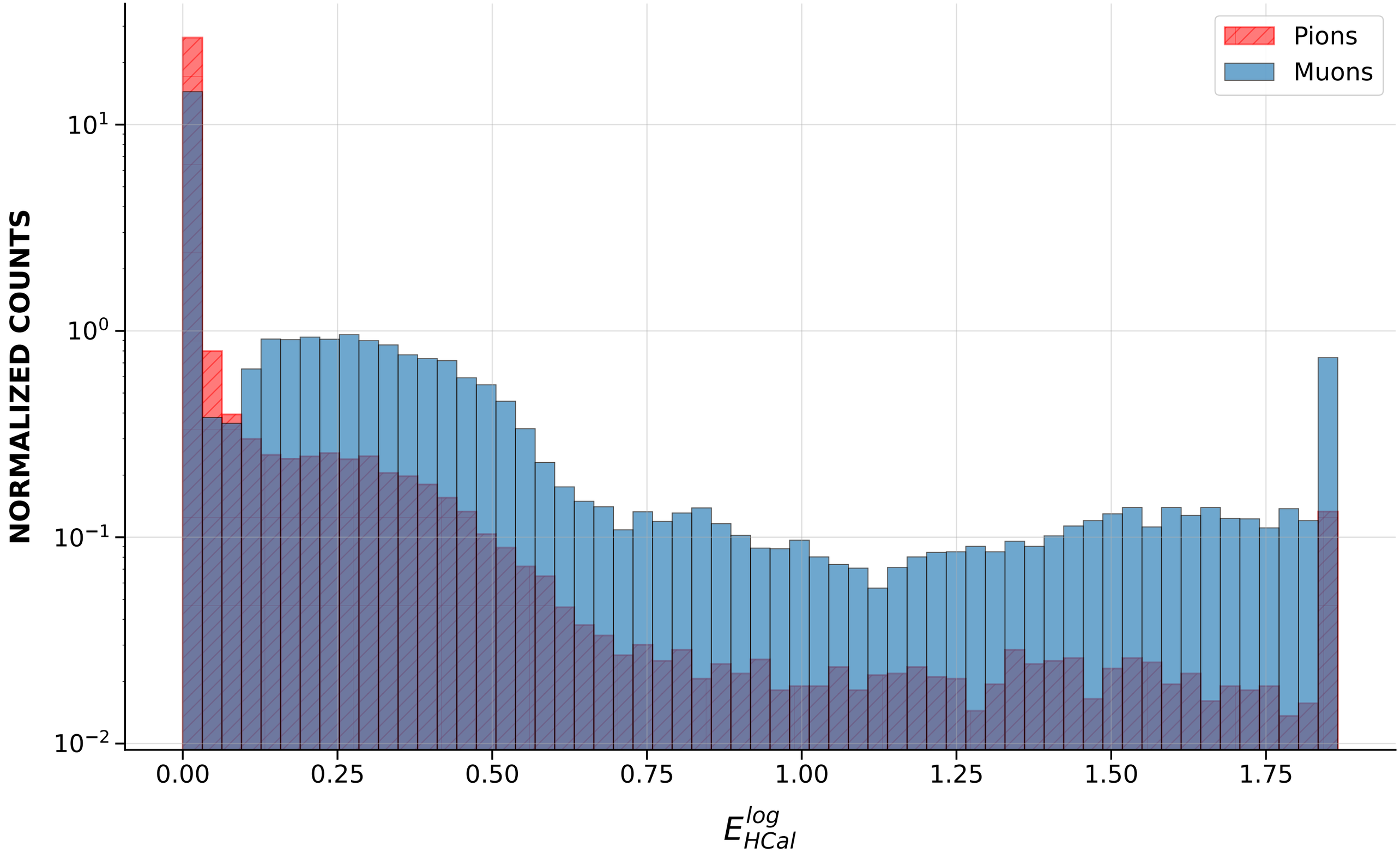
ENGINEERED CALORIMETER FEATURE
EoverP_ratio



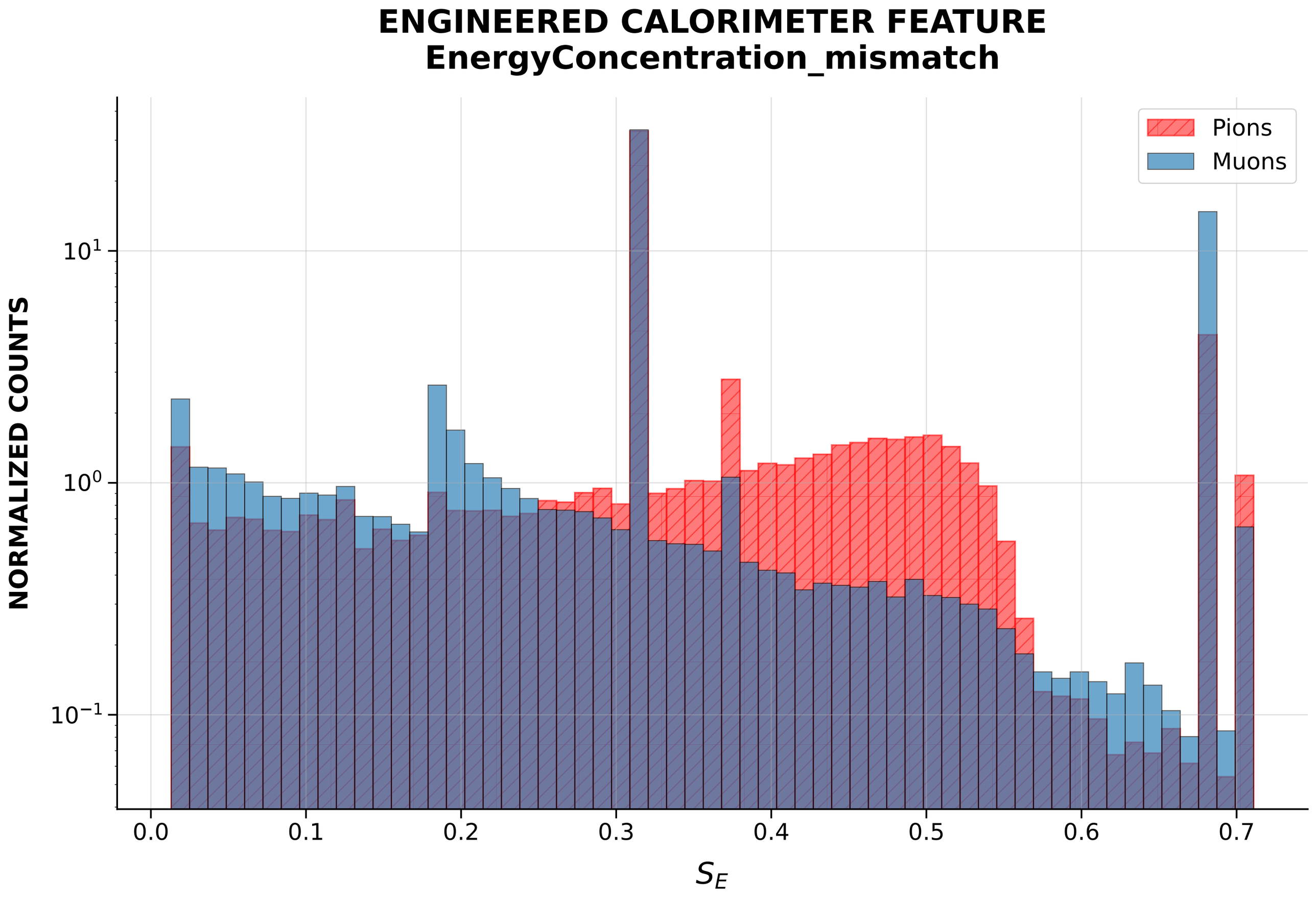
ENGINEERED CALORIMETER FEATURE
logECal



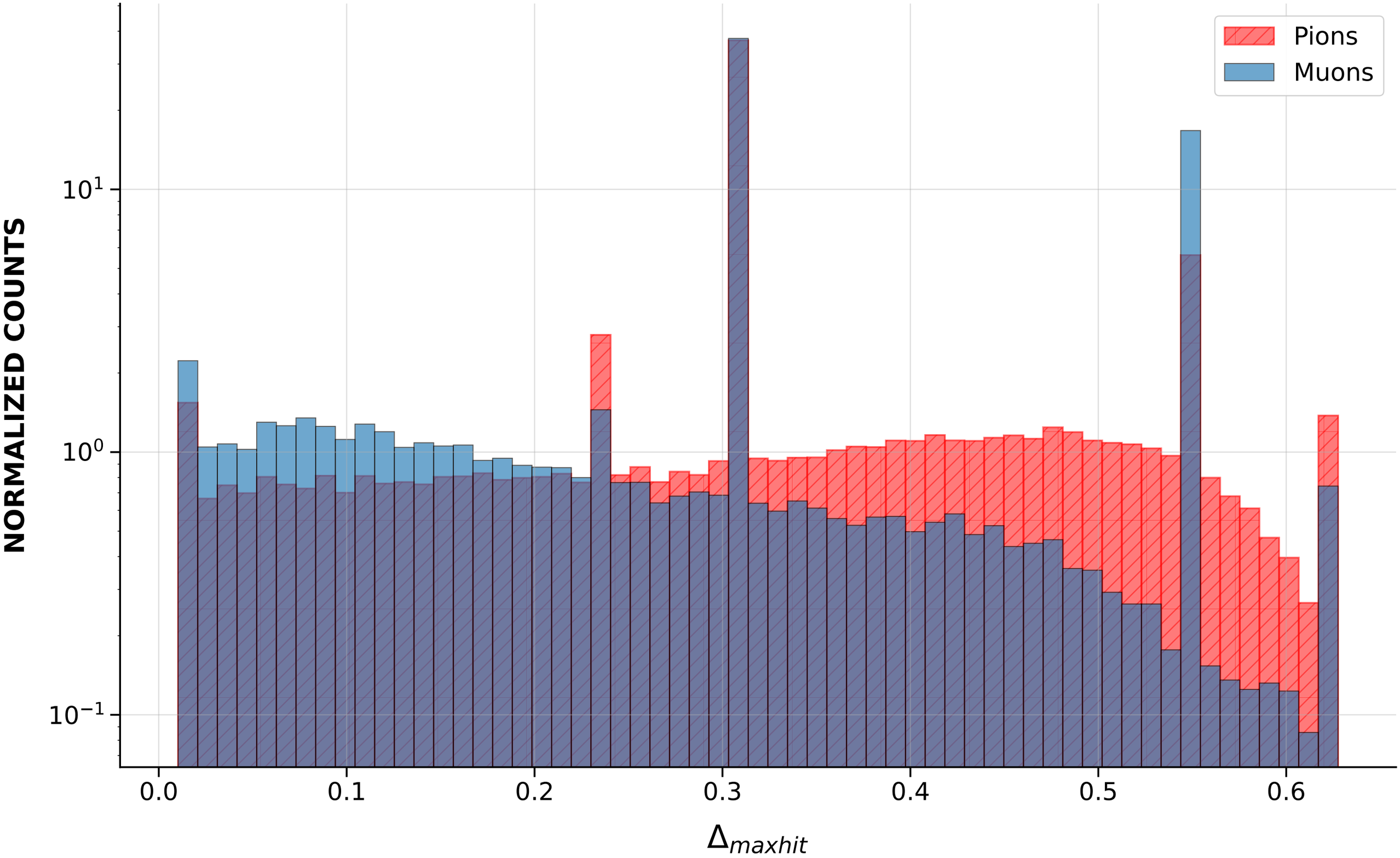
ENGINEERED CALORIMETER FEATURE
logHCal



ENGINEERED CALORIMETER FEATURE
EnergyConcentration_mismatch



ENGINEERED CALORIMETER FEATURE
MaxHitFrac_mismatch



ENGINEERED CALORIMETER FEATURE
HitRatio

